SUMMARY

- The current, I flowing in a conductor $I = \frac{dQ}{dt}$, where dQ is the charge that flows through a cross-section in a time interval dt. SI unit of current is ampere (A). 1A = 1 C s^{-1}.
- The current density J in a conductor is the current flowing per unit area. $\left(J = \frac{I}{A}\right)$
- Current is a scalar but current density is a vector.
- The general form of Ohm's law $\vec{J} = \sigma\vec{E}$
- Practical form of Ohm's law states that $V \propto I$, or $V = IR$ where I is the current and R is the resistance, V potential difference between ends of the conductor.
- The resistance R of a conductor is $R = \frac{V}{I}$. SI unit of resistance is ohm (Ω) and $1\ \Omega = \frac{1V}{1A}$
- The resistance of a conductor $R = \rho\frac{l}{A}$ where l is length of the conductor and A is its area of cross section.
- The resistivity of a material determines how much resistance it offers to the flow of current.
- The equivalent resistance (R_S) of several resistances (R_1, R_2, R_3……..) connected in series is $R_S = (R_1 + R_2 + R_3$……..)
- The equivalent resistance (R_P) of several resistances (R_1, R_2, R_3……..) connected in parallel is $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ...$
- Kirchhoff's first rule (Current rule or junction rule): The algebraic sum of the currents at any junction is zero.
- Kirchhoff's second rule (Voltage rule or loop rule): In a closed circuit the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit.
- Electric power is the rate at which electric energy is delivered.
- If a current I flows across a potential difference V, the power delivered to the circuit is $P = IV$.
- In a resistor R, the electrical power converted to heat is $P = I^2R = \frac{V^2}{R}$
- The energy equivalent of one kilowatt-hour (kWh) is 1kWh = 3.6 X 10^6 J.
- Metre bridge is one form of Wheatstone's network.
- Potentiometer is used to compare emf of cells.
- Joule's law of heating is $H = VIt$ (or) $H = I^2Rt$.

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